

Complexity-Based Market Regime Detection for Dynamic Asset Allocation

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Abstract

This project develops a complexity-based market regime detection framework for dynamic asset allocation between NIFTYBEES and GILT5YBEES. Market behaviour is characterized using four complexity measures: Hurst Exponent, Largest Lyapunov Exponent, Sample Entropy, and Lag-1 Autocorrelation. These metrics are used to classify the market into three regimes: Equilibrium, Complete Chaos, and Edge-of-Chaos (Momentum); which drive portfolio allocation decisions. A realistic transaction cost model is incorporated to improve practical applicability. The framework is calibrated using historical data from 2014–2022 and evaluated through an out-of-sample walk-forward test from 2023–2026. Results indicate that the regime-aware strategy outperformed a passive NIFTYBEES benchmark, achieving higher risk-adjusted returns and lower drawdowns. The findings suggest that complexity metrics can provide useful information about changing market conditions and support more resilient portfolio allocation decisions.

1. Introduction

Financial markets are not stationary. Extended periods of trending, persistent behavior are routinely interrupted by episodes of heightened uncertainty, volatility clustering, and structural breakdown. Traditional portfolio construction frameworks such as Mean-Variance Optimization and static factor models do not account for these shifting regimes and are consequently vulnerable to sudden shocks, larger drawdowns, and degraded out-of-sample performance.

This paper develops a complexity-based market regime detection framework for dynamic asset allocation. Rather than modeling return moments directly, the approach characterizes the underlying geometry and computational structure of market dynamics using four nonlinear invariants: the Hurst Exponent, Largest Lyapunov Exponent, Sample Entropy, and Lag-1 Autocorrelation. These metrics jointly form a multidimensional state vector that captures the prevailing market environment in real time.

A key departure from conventional regime models is the classification scheme. Standard approaches label regimes as bull, bear, or sideways; these descriptions are directional but not actionable for risk management. This framework instead classifies regimes according to their dynamical structure: Equilibrium (mean-reverting, low complexity), Complete Chaos (high entropy, weak predictability), and Edge-of-Chaos (persistent, momentum-sustaining). This taxonomy maps directly to portfolio allocation decisions, making the output immediately useful rather than requiring a secondary translation step.

The framework is applied to two NSE-listed exchange-traded funds: NIFTYBEES, which tracks the Nifty 50 Index and provides equity market exposure, and GILT5YBEES, a 5-year government securities ETF serving as the defensive asset. The primary objective is to improve risk-adjusted performance and reduce drawdowns relative to a passive buy-and-hold benchmark through regime-aware capital allocation

2. Methodology

Complexity Metrics

Market behavior is represented using four complementary metrics, each computed on rolling windows of daily log returns of NIFTYBEES. The input to all metrics is the continuously compounded log return series, $r_t = \ln(P_t / P_{t-1})$, which provides time-additive properties suitable for dynamical analysis.

Hurst Exponent (H)

The Hurst Exponent quantifies the long-range dependence and fractional dimensionality of the return series, estimated via Rescaled Range (R/S) analysis. For a return series partitioned into sub-windows of length n , the R/S statistic scales asymptotically as:

$$E[R(n)/S(n)] = c \cdot n^H$$

where H is extracted as the slope of the log-log regression of $E[R/S]$ against n .

The value of H characterizes the geometric structure of the price process:

$H > 0.5$ indicates fractional persistence, where positive covariance between disjoint increments sustains self-reinforcing trends;

$H \approx 0.5$ indicates a memoryless random walk; and

$H < 0.5$ indicates anti-persistence and mean-reversion.

Largest Lyapunov Exponent (LLE)

The Largest Lyapunov Exponent measures the mean exponential rate at which infinitesimally close trajectories diverge in a reconstructed phase space; which is a direct measure of sensitive dependence on initial conditions. A large positive LLE indicates rapid contamination of the phase space by small perturbations, signaling the collapse of the predictability horizon. A value near or below zero indicates convergence to a stable or periodic trajectory.

Sample Entropy (SampEn)

Sample Entropy quantifies the sequential irregularity and algorithmic complexity of the return series. It is defined as the negative natural logarithm of the conditional probability that two sequences matching for m consecutive points will continue to match at $m+1$ points:

$$\text{SampEn}(m, r, N) = -\ln(C^{(m+1)}(r) / C^m(r))$$

where $C^m(r)$ is the total count of template matches across the phase space at embedding dimension m and tolerance radius r . SampEn is preferred over the closely related Approximate

Entropy (ApEn) because it is free from self-counting bias and remains well-defined for the moderate sample sizes characteristic of rolling financial windows.
High SampEn reflects maximal disorder and unpredictability.

Lag-1 Autocorrelation (Rho)

The lag-1 autocorrelation provides a linear baseline for short-term return memory:

$$\text{Rho} = \text{Cov}(r_t, r_{t-1}) / \text{Var}(r_t)$$

It functions as a fast-moving signal that captures immediate shifts in market microstructure. Strongly negative values at daily frequency can reflect microstructure artifacts such as bid-ask bounce rather than genuine mean-reversion, a distinction that is accounted for in the classification framework.

These metrics jointly form a multidimensional market state representation:

$$Z_t = [\text{Hurst Exponent}_t, \\ \text{Largest Lyapunov Exponent}_t, \\ \text{Sample Entropy}_t, \\ \text{Lag-1 Autocorrelation}_t]$$

which characterizes the prevailing market environment.

3. Regime Classification Framework

Theoretical Grounding: Wolfram's Computational Universality

The classification framework is grounded in Stephen Wolfram's theory of computational universality, developed in *A New Kind of Science* (2002). Wolfram demonstrated that all sufficiently complex dynamical systems, regardless of their physical substrate, can be classified into four universal behavioral classes based on their long-run computational properties. This project applies that taxonomy to financial markets, treating price dynamics as the output of a complex computational system whose behavioral class can be identified from observable invariants.

This project maps four classes to distinct market environments as follows. Class 1 and Class 2 systems converge to fixed points or stable periodic orbits, the financial analog is a mean-reverting, equilibrium market with bounded fluctuations and low informational complexity. Class 3 systems produce aperiodic, effectively random output, the financial analog is a market in complete chaos, where the predictability horizon has collapsed and historical price structure offers no systematic edge. Class 4 systems operate at the boundary between order and chaos, exhibiting persistent, structured, information-rich dynamics, the financial analog is a trending market in which momentum is self-sustaining and macroeconomic information propagates through prices in an organized way.

Langton (1990) established that Class 4 systems possess the maximal capacity to store and transmit structural information. This is the precise mathematical substrate required for the emergence and persistence of momentum in financial markets. The implication is direct: momentum strategies should be expected to work in Class 4 environments and fail in Class 3 environments, not because of investor psychology, but because of the underlying computational properties of the system.

Why These Four Metrics

Each of the four dynamical invariants in the state vector Z_t probes a distinct and complementary dimension of the market's computational class.

The Hurst Exponent probes the fractal dimension and long-range memory of the return series. It answers the question of whether the market is operating as a persistent, self-reinforcing system ($H > 0.5$) or a mean-reverting one ($H < 0.5$). This directly distinguishes Class 4 from Class 1/2. However, H alone cannot distinguish Class 3 from Class 4: both can exhibit H near 0.5, but through entirely different mechanisms — one from genuine randomness, the other from the transition between persistence and chaos.

The Largest Lyapunov Exponent probes the stability of the phase space. It answers the question of whether nearby market trajectories remain close over time or diverge exponentially. A large positive LLE is the defining signature of Class 3 dynamics; explosive divergence that destroys the predictability of any model, however sophisticated. This is the metric that most directly captures the onset of a crisis regime before it fully manifests in realized volatility, because trajectory divergence usually precedes the volatility spike in the return data.

Sample Entropy probes the informational complexity of the return sequence. It answers the question of whether the market is generating new, unpredictable information (high entropy, Class 3) or operating with structured, recurring patterns (moderate entropy, Class 4) or near-deterministic repetition (low entropy, Class 1/2). SampEn is particularly valuable because it is sensitive to the texture of market dynamics rather than just their magnitude; two markets with identical volatility can have very different entropy profiles.

Lag-1 Autocorrelation probes the linear memory structure of returns at the shortest timescale. It provides a fast-moving confirmation signal that complements the slower-moving invariants. Positive autocorrelation is consistent with Class 4 momentum dynamics; near-zero autocorrelation is consistent with Class 3 randomness; mildly negative autocorrelation is consistent with Class 1/2 mean-reversion.

The Three Regimes

The state vector is mapped to three economically interpretable regimes through a proprietary scoring and classification mechanism calibrated entirely on in-sample data.

State 0: Equilibrium. The market exhibits low complexity, mild mean-reversion, and bounded fluctuations. The dynamical invariants collectively indicate a Class 1/2 environment: low long-range dependence, stable trajectories, low informational entropy, and slightly negative short-term autocorrelation. This regime does not offer a systematic directional edge.

State 1: Complete Chaos. The market exhibits high entropy, elevated trajectory divergence, and weak or collapsed predictability. The dynamical invariants indicate a Class 3 environment where historical price structure has ceased to be informative. Critically, this regime is detected not by waiting for a realized volatility spike but by identifying the preceding geometric collapse; the point at which the market's computational class shifts from structured to random.

State 2: Edge of Chaos / Momentum. The market exhibits persistent, information-rich dynamics with structured trend continuation. The dynamical invariants indicate a Class 4 environment: elevated long-range dependence, moderate but not explosive trajectory divergence, moderate informational entropy, and positive short-term autocorrelation.

Persistence Filter

A minimum-duration filter is applied to prevent excessive regime switching driven by short-term estimation noise. Any detected regime block shorter than a minimum threshold is merged into the preceding stable regime. This enforces an economic constraint: genuine macroeconomic regime transitions persist for weeks to months, not days. The filter reduces turnover costs and prevents the allocation framework from reacting to transient metric fluctuations that do not represent true structural shifts.

In-sample classification of regime is given in Fig. 1 below.

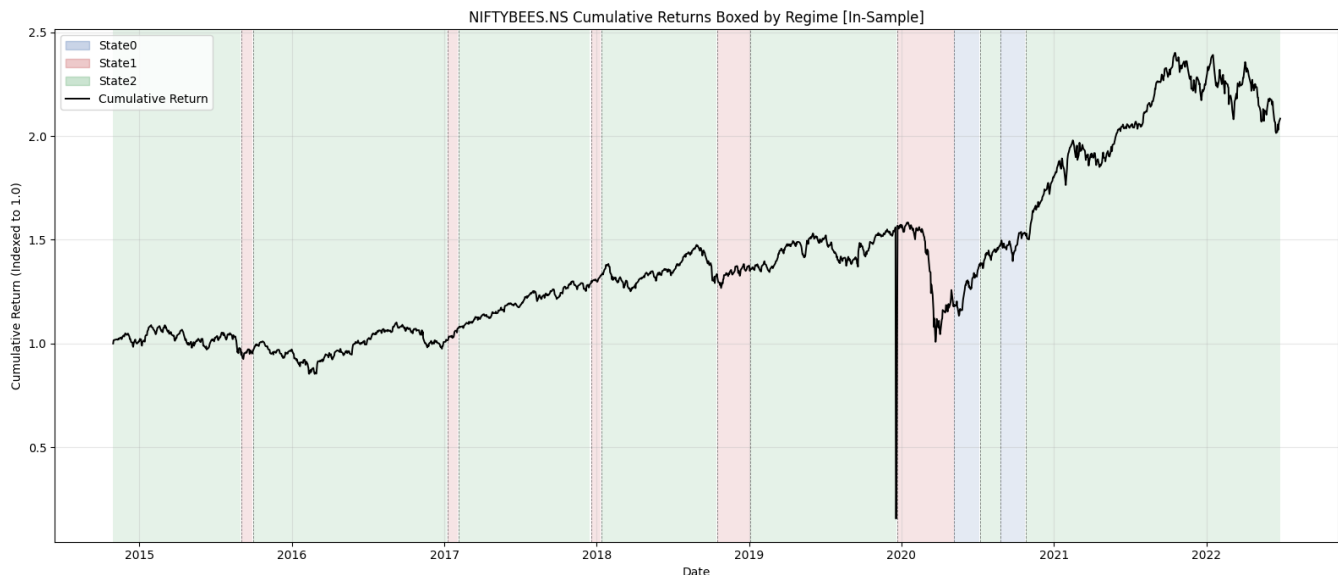


Fig. 1: In-Sample results from Nifty BEES (Jan. 2014 till Jun. 2022) is used to build, test and calibrate the model parameters. In order to avoid overfitting, some degree of error in regime detection was tolerated.

In-sample evaluation of the model during some important market events, where values are represented as percentage of days classified within a regime:

Event	% days in State 0	% days in State 1	% days in State 2
Covid (Feb. – Apr. 2020)	0.00%	100.00%	0.00%
Bull market of 2017 (Jan. 2017 – Dec. 2017)	0.00%	12.20%	87.80%
IL&FS (Sept. 2019 – Dec. 2019)	0.00%	69.20%	30.80%
Calm 2019 (Jan. 2019 – Jun. 2019)	0.00%	0.00%	100.00%

Note: The 2019 IL&FS crisis was primarily a credit and liquidity event; its impact on the equity market materialized as a sustained directional drawdown rather than an acute volatility spike. Consequently, the market exhibited a high Hurst Exponent (indicating strong persistence of a directional move), a moderate Largest Lyapunov Exponent (demonstrating that phase-space trajectories were not diverging explosively), a moderate Sample Entropy (reflecting a lack of sudden structural disorder), and a positive lag-1 autocorrelation (confirming short-term trend continuation). By definition, this dynamical signature satisfies the structural parameters of a State 2 (Edge-of-Chaos) regime, manifesting empirically as sustained downward momentum rather than a sudden panic.

4. Portfolio Construction and Implementation Mechanics

The detected market regime directly dictates the capital allocation between the equity leg (NIFTYBEES) and the defensive bond leg (GILT5YBEES). The portfolio is constructed as a fully invested, unlevered strategy that incorporates regime-dependent asset allocation, volatility targeting, position constraints, and realistic transaction cost drags.

A. Calibration and Out-of-Sample Design

The model parameters are calibrated exclusively using historical in-sample data from January 2014 to June 2022. To strictly mitigate information leakage and look-ahead bias, a six-month embargo period spanning from July 2022 to December 2022 is excluded from both the training and testing phases. The final evaluation is conducted via a walk-forward out-of-sample test from January 2023 to May 2026.

B. Regime Allocation Bounds and Volatility Targeting

The allocation framework decomposes the weighting mechanism into two distinct decisions: (1) determining the strategic tier warranted by the current classified regime, and (2) sizing the exposure within that tier based on current market volatility. Portfolio volatility is dynamically scaled within regime-conditional bounds to ensure targeted risk parameters are met while maintaining a fully invested posture.

C. Transaction Cost Model

To ensure the backtest reflects real-world implementability, explicit delivery-based National Stock Exchange (NSE) transaction costs are applied symmetrically at every rebalancing event. Because the portfolio is fully invested, increasing equity exposure necessitates selling bonds, with both legs of the trade incurring respective friction costs.

Transaction Cost Assumptions (2026 NSE Rates):

Cost Component	Equity Buy	Equity Sell	Bond Buy	Bond Sell
STT	0.10%	0.10%	0.00%	0.00%
Stamp Duty	0.015%	0.00%	0.015%	0.00%
Exchange + GST	~0.0035%	~0.0035%	~0.0035%	~0.0035%
Total Drag	~11.85 bps	~10.35 bps	~1.85 bps	~0.35 bps

Note on Taxation and Brokerage: Securities Transaction Tax (STT) is not levied on debt-oriented ETFs such as GILT5YBEES. Brokerage fees are assumed to be zero, aligning with standard discount broker structures for delivery trades. Furthermore, Capital Gains Tax (CGT) is excluded from the model. This is consistent with standard quantitative backtesting practices, as CGT liabilities are heavily dependent on investor-specific variables such as individual tax slabs, holding periods, and prior-year offsetable losses.

5. Results and Performance Analysis

The proposed regime-aware dynamic allocation strategy was empirically evaluated via a walk-forward test and compared against a passive buy-and-hold strategy for NIFTYBEES. The strategy's performance, computed net of all modelled transaction costs, demonstrated substantial improvements in both absolute and risk-adjusted metrics.

Out-of-Sample Performance Summary (Jan 2023 – May 2026):

Performance Metric	Regime-Aware Strategy	Buy & Hold (NIFTYBEES)
Annualized Return	13.53%	9.62%
Annualized Sharpe Ratio	1.33	0.83
Maximum Drawdown	-10.51%	-15.23%
Calmar Ratio	1.29	0.63

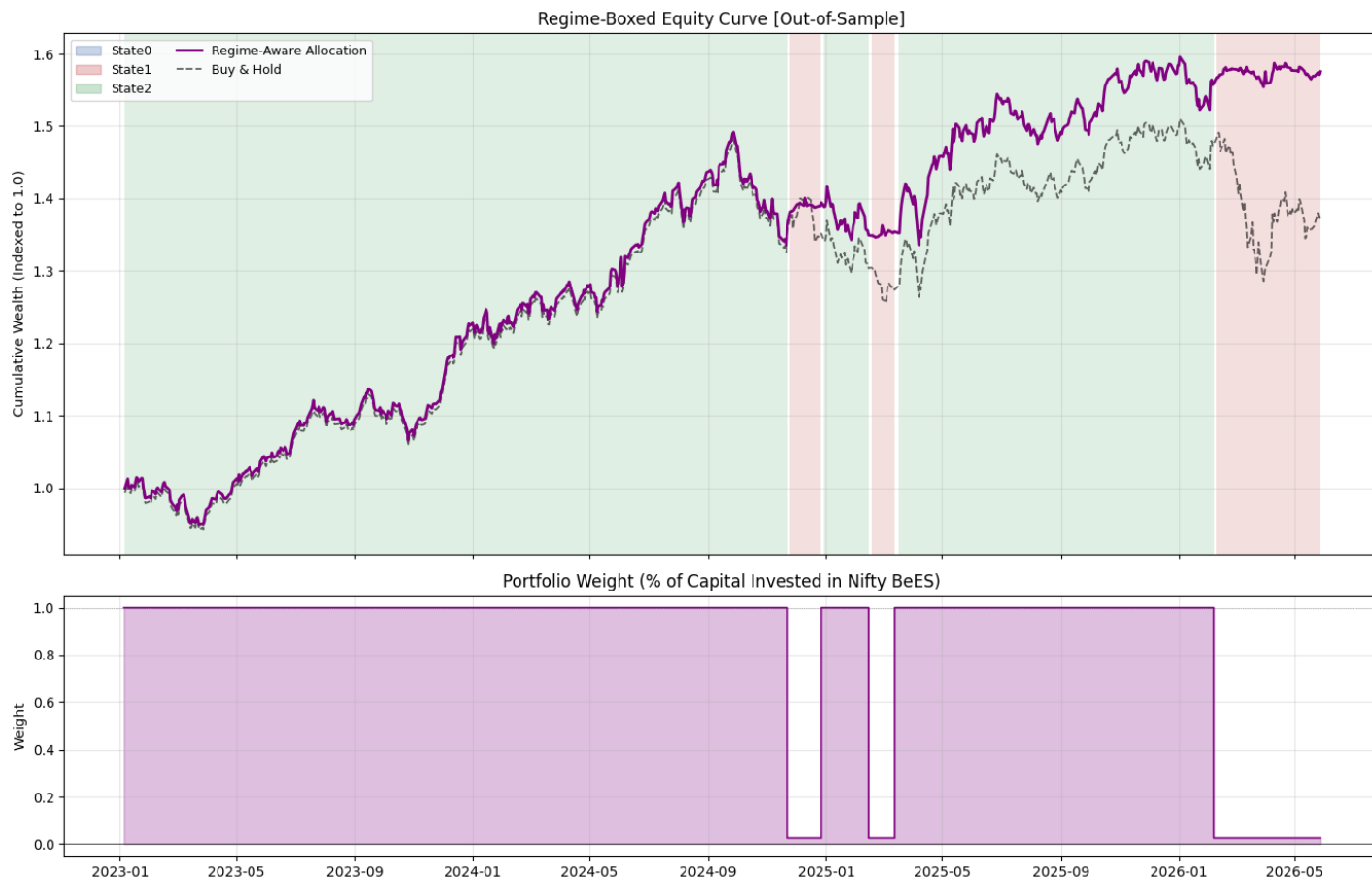


Fig. 2: (Top) Out-of-sample walk-forward test results comparing the cumulative wealth of the regime-aware portfolio against the benchmark between January 2023 and May 2026. (Bottom) Corresponding dynamic weight allocation to NIFTYBEES.

Key Findings

Alpha Generation: The regime-aware strategy achieved a significant improvement in profitability, generating an excess annualized return of 3.91% over the passive benchmark, even after accounting for transaction cost drag.

Risk-Adjusted Outperformance: The integration of regime-conditional allocation bounds and a daily volatility target of 1.2% led to vastly superior risk-adjusted returns. This is reflected in an annualized Sharpe Ratio of 1.33, outperforming the buy-and-hold strategy by 0.50.

Downside Protection: The magnitude of the maximum drawdown was successfully reduced by 4.72 percentage points (from -15.23% to -10.51%). The Calmar Ratio effectively doubled, highlighting the model's efficiency in managing tail risk relative to its return.

Dynamic Exposure Management: The largest performance benefits arose from active downside protection during unstable market environments. The strategy successfully mitigated risk by autonomously reducing NIFTYBEES exposure during detected "State 1" (Complete Chaos) regimes, which historically exhibit severe volatility and structural breakdowns.

Unobserved State 0 Dynamics: The model did not classify any periods as State 0 (Equilibrium) during the out-of-sample testing phase. Therefore, the strategy's operational efficacy and risk-adjusted performance during prolonged mean-reverting environments cannot be definitively evaluated from this specific test window.

Ultimately, these out-of-sample results validate that complexity metrics contain highly actionable information regarding latent shifts in market structure, providing a robust foundation for constructing resilient, regime-adaptive portfolios.

6. Conclusion

This project demonstrates that nonlinear complexity measures can be effectively integrated into a practical market-regime detection framework for quantitative portfolio management. By synthesizing the Hurst Exponent, Largest Lyapunov Exponent, Sample Entropy, and Lag-1 Autocorrelation, the model successfully identifies persistent, chaotic, and equilibrium market environments, and adjusts portfolio allocations accordingly.

The rigorous out-of-sample walk-forward results indicate that regime-aware allocation can significantly improve risk-adjusted performance while substantially mitigating drawdowns relative to a passive investment approach. Ultimately, this framework provides a promising, theoretically grounded foundation for future research involving the application of complex dynamics to advanced portfolio construction techniques.

7. Limitations and Future Work

While the proposed framework demonstrates robust out-of-sample performance, certain methodological limitations remain. First, the regime classification boundaries and the minimum-duration persistence filter are calibrated using in-sample empirical quantiles, which introduces a degree of sample dependence into the model. Second, state transitions are assigned via a discrete, rule-based scoring mechanism rather than being estimated probabilistically, meaning the certainty of a regime shift is not mathematically quantified. Finally, the framework does not currently incorporate formal statistical significance testing for the detected transitions between the computational classes.

Future research should address these constraints by investigating the integration of continuous probabilistic models.